

Qatar-1b

R-2514

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110603 · created 2026-08-25T00:33:04+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455715.79486 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.015
Transit depth (Rp/Rs)^2	2.36 +/- 0.47 [%]
Orbital Inclination (inc)	84.16 +/- 0.81
Airmass coefficient 1 (a1)	0.998 +/- 0.013
Airmass coefficient 2 (a2)	0.0029 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	#2 - [533, 50]
Optimal Aperture	3.44
Optimal Annulus	14.19
Transit Duration (day)	0.0695 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.015 vs 0.1463 (deviation 0.51 σ)
Duration fit vs published	0.0695 d vs 0.0692 d (deviation 0.07 σ)
Mid-time O-C	1.0 min
Depth SNR	10.3
Residual scatter	1.28 %

Light curve

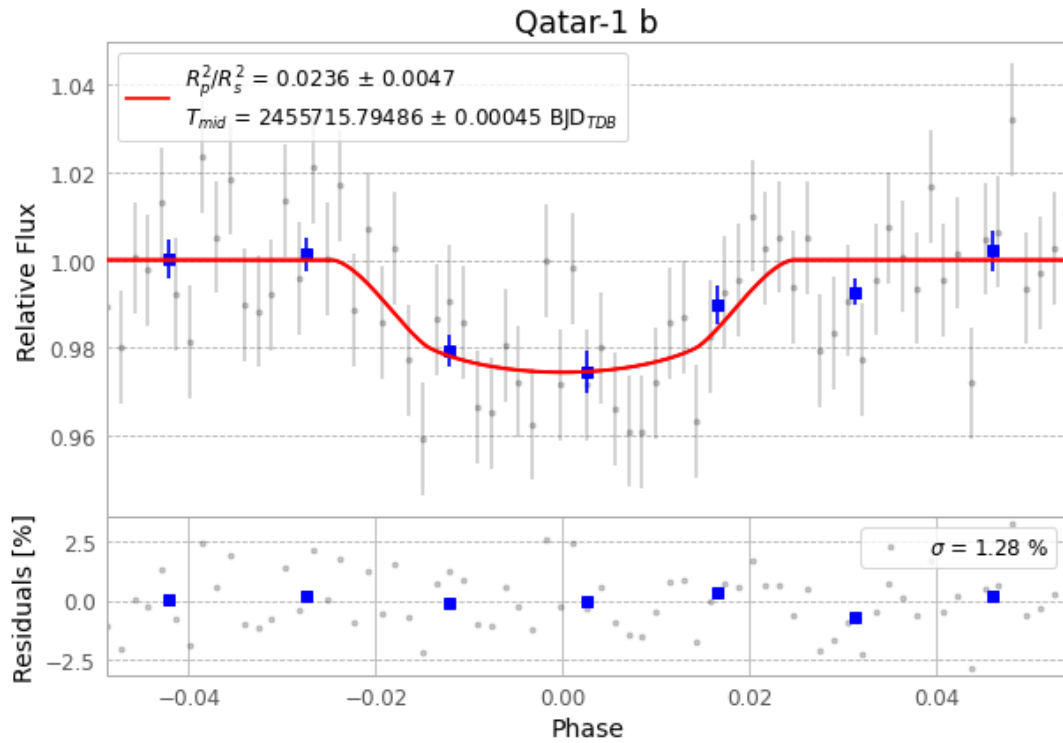


Figure 1 — FinalLightCurve_Qatar-1 b_2011-06-02.png (SHA-256 2dc28ba6cbe074a0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ac516de22490829a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), Qatar-1110603052412.FITS → Qatar-1110603085413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110603.log (46f793d32894...), FinalLightCurve_Qatar-1 b_2011-06-02.png (2dc28ba6cbe0...), FinalLightCurve_Qatar-1 b_2011-06-02.csv (b373f20ec98c...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:33Z.